Yanghao Chen

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Education

Tongji University — Shanghai, China
Bachelor of Engineering in Engineering Mechanics — Sep. 2023 – Present; expected Jun. 2027
GPA: 87/100

Johns Hopkins University — Baltimore, MD, USA
Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering — Jan. 2026 – Oct. 2026

Research Interests: Computational Mechanics, Scientific Machine Learning, Differentiable Programming

Research Projects

GPU-accelerated Hybrid Finite Element–Neural Operator Solver (Ongoing)
Jan. 2026 – Present
Lead Undergraduate Researcher | Advisor: Prof. Somdatta Goswami — *Johns Hopkins University*

- Investigating a hybrid finite element–neural operator solver using non-overlapping domain decomposition, coupling FEM and neural-operator subdomains through iterative exchange of interface displacements and reaction forces.
- Generated interface displacement samples using Gaussian random fields and computed the corresponding subdomain stress fields with GPU-accelerated JAX-FEM simulations. Trained Transolver in JAX on these paired data to learn the mapping from interface displacement fields to subdomain stress fields.
- Investigated convergence acceleration strategies and relaxation parameters for FEM–neural operator coupling. Currently extending the solver to 3D fiber-reinforced composites, with neural-operator weights shared across fiber subdomains.

PINNs for Shock Capturing in the Burgers Equation
Nov. 2025 – Mar. 2026
Undergraduate Researcher | Advisor: Prof. Xianyang (Tom) Chen — *Tongji University*

- Trained a PINN in PyTorch for the one-dimensional inviscid Burgers shock problem, using PDE-residual, initial-condition, and boundary-condition losses with automatic differentiation.
- Added an artificial-viscosity term with a fixed coefficient to the PDE residual and trained a second PINN to predict the shock profile.
- Compared both PINNs with the analytical inviscid solution, examining shock location and transition width; observed that fixed artificial viscosity reduced the relative L^2 error while increasing the maximum pointwise error.

ConvLSTM Prediction of Concentration and Stress Fields in Battery Materials
Jun. 2025 – Oct. 2025
Undergraduate Researcher | Advisor: Prof. Ying Zhao — *Tongji University*

- Generated polycrystalline NMC microstructures and automated MATLAB–COMSOL simulations to produce time-aligned image sequences of lithium concentration and von Mises stress.
- Trained separate conditional ConvLSTM models in PyTorch using past field sequences, grain-orientation maps, and C-rate inputs, with MSE and SSIM losses and scheduled sampling to introduce model-generated inputs during training.
- Evaluated autoregressive concentration and stress predictions against COMSOL reference sequences, examining error accumulation over successive prediction steps.

Domain-Specific LLM Fine-Tuning for Mechanics of Materials
Sep. 2024 – Apr. 2025
Undergraduate Researcher | Advisor: Prof. Ying Zhao — *Tongji University*

- Built instruction–response datasets from textbooks and technical literature, covering stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B with LoRA on Google Colab for mechanics of materials Q&A and concept explanations.
- Released two instruction datasets and two fine-tuned Qwen2.5-based checkpoints on Hugging Face.

Technical Skills

Scientific Machine Learning	PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM
Computational Mechanics	JAX-FEM, FEniCSx, Abaqus, COMSOL Multiphysics (LiveLink for MATLAB), Gmsh
Programming & Tools	Python, MATLAB, Linux, Git/GitHub, Matplotlib, Origin
LLM Fine-Tuning	Qwen2.5, LoRA, Instruction Dataset Curation
Languages	Mandarin Chinese (native), Cantonese, English